UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 009 · Introduction to maxima and minima for a function of two variables · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S009·7	Deck 009, slide 7 (rendered .pptx)
T009 00:07:04	AV-summary transcript 009 (“Extremas of functions”), segment start
CW U1 P31	Unit 1 class-work problem 31
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c

Contents

1. Figure G1.1 · Local maxima: peaks with horizontal tangent planes
2. Figure G1.2 · A local minimum: $z = x^2 + y^2$
3. Figure G1.3 · A saddle point: $z = x^2 - y^2$
4. Figure G1.4 · One-variable picture: maxima, minima and $\frac{dy}{dx} = 0$

Colours follow the Okabe–Ito order: blue = the surface or curve, vermilion = the stationary point and its tangent plane, green = second trace. Figures G1.1–G1.3 redraw the deck's pictures of maxima, minima and a saddle with the illustrative functions $e^{-x^2-y^2}$-type bumps, $x^2 + y^2$ and $x^2 - y^2$; the slides' screenshots (including a photograph of a horse saddle) are not copied.

1 Maxima, minima and saddle points

two local maxima (peaks)

vermilion: horizontal tangent plane

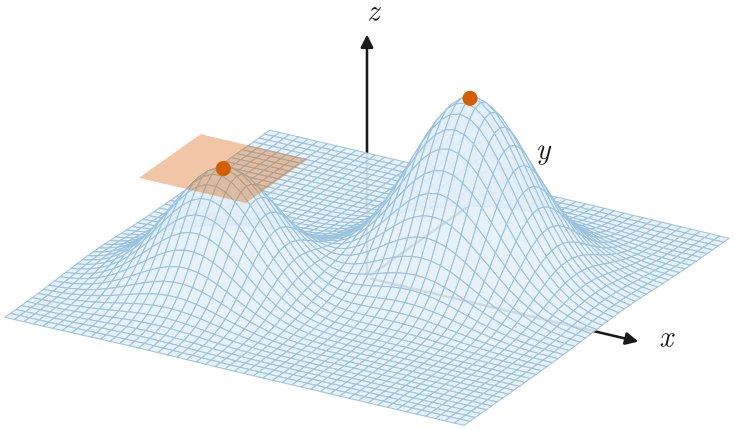


Figure G1.1. Local maxima: peaks with horizontal tangent planes. ESA ★

S009·3 · T009 00:00:16

How to read it. A surface with two peaks (illustrative: a sum of two bell shapes). Both peaks are local maxima; the taller one is also the largest value overall. The vermilion square is the horizontal tangent plane at the lower peak.

$z = x^2 + y^2$

minimum at $(0,0)$: $rt - s^2 = 4 > 0,\ r = 2 > 0$

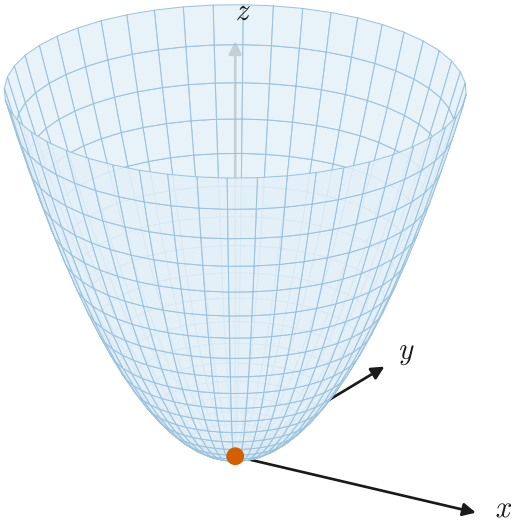


Figure G1.2. A local minimum: $z = x^2 + y^2$. ESA ★

S009·4 · T009 00:02:11

How to read it. The bowl $z = x^2 + y^2$ has its lowest point at the origin. Every nearby point is higher: $\Delta > 0$. Its test values are $r = t = 2$, $s = 0$, $rt - s^2 = 4 > 0$, $r > 0$ (W3.1).

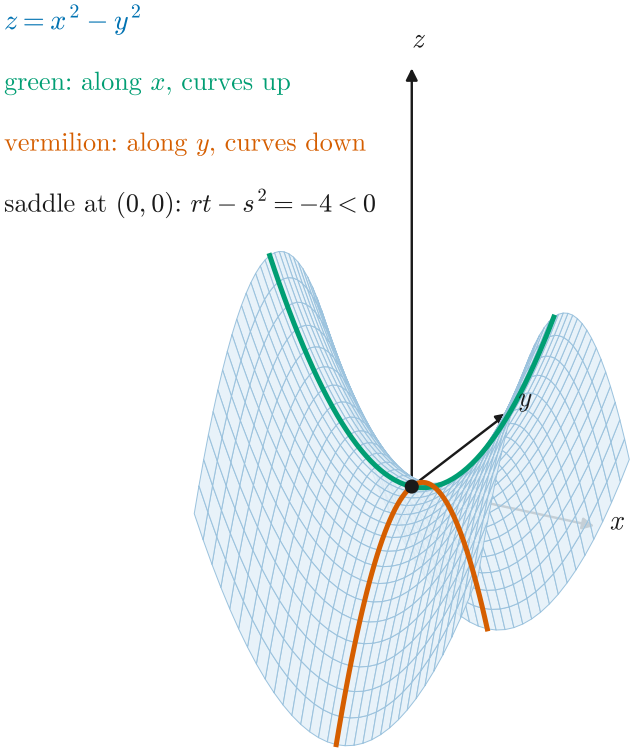


Figure G1.3. A saddle point: $z = x^2 - y^2$. ESA ★★

S009·5 · S009·7

How to read it. Along the x-direction (green) the surface curves up, along the y-direction (vermilion) it curves down. The origin is stationary but neither a maximum nor a minimum; $rt - s^2 = -4 < 0$ (W3.2).

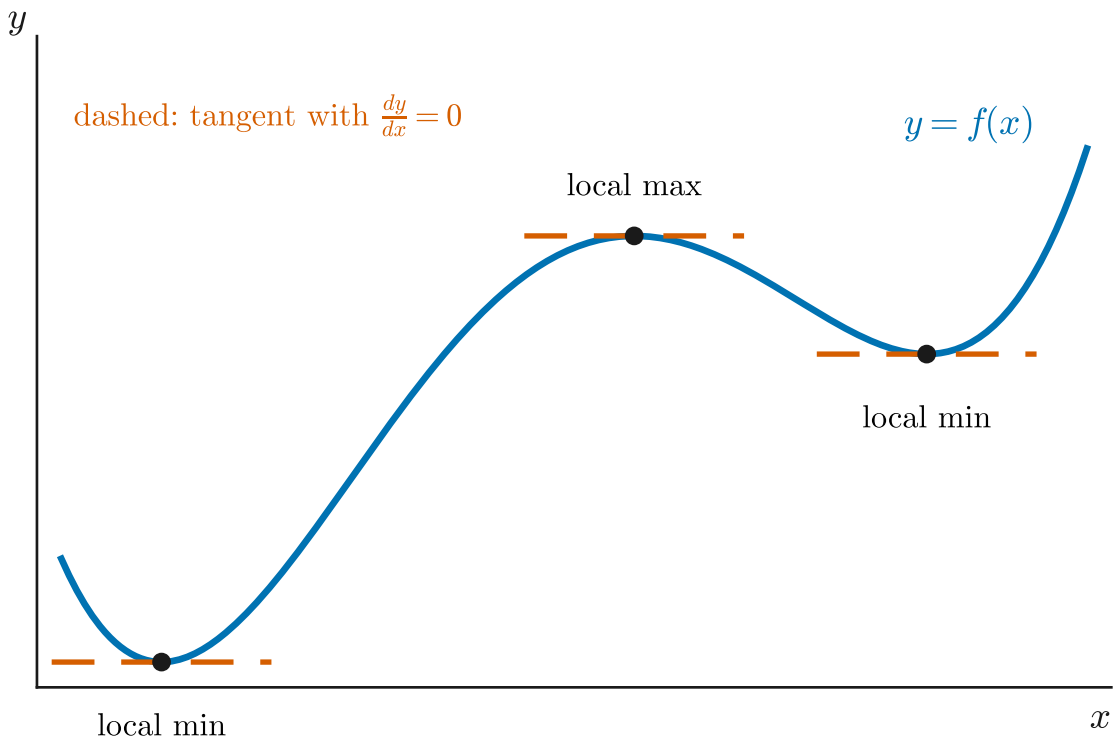


Figure G1.4. One-variable picture: maxima, minima and $\frac{dy}{dx} = 0$. ESA ★

T009 00:00:50 · T009 00:01:10

How to read it. On $y = f(x)$ the tangent is horizontal (slope 0) at every local maximum and minimum (short dashed segments). The two-variable test copies this: $f_x = f_y = 0$ replaces $\frac{dy}{dx} = 0$, and $rt - s^2$ with r replaces $\frac{d^2y}{dx^2}$.